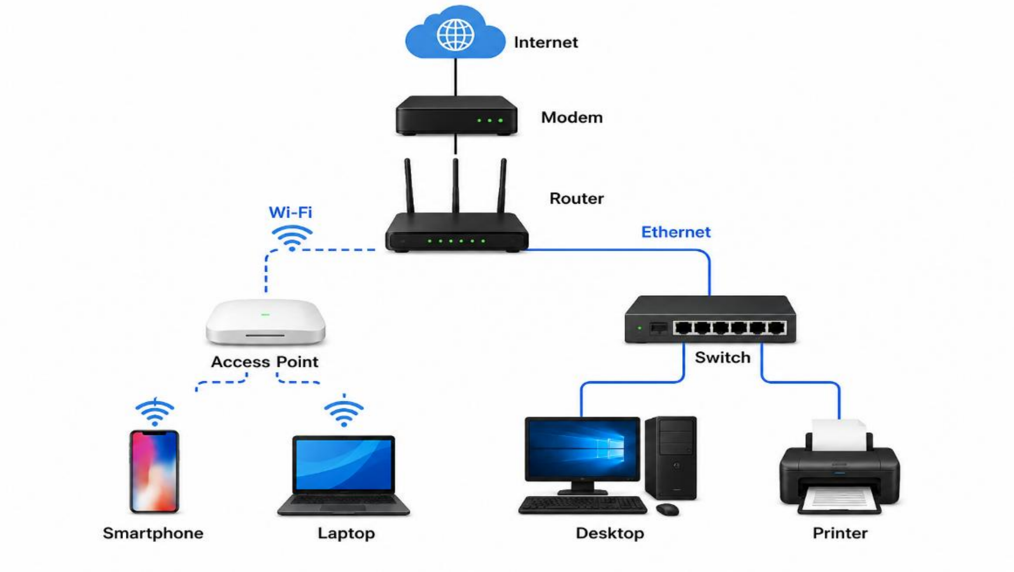
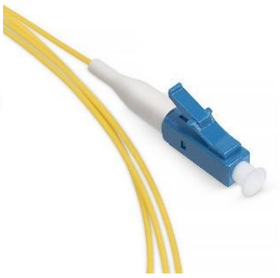
1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Connects your home network to the Internet Service Provider (ISP).	Installed at home by Airtel, JioFiber, or BSNL to provide internet access.
Router	Shares the internet connection among multiple devices and assigns IP addresses.	Home Wi-Fi router connecting phones, laptops, and TVs.
Switch	Connects multiple wired devices within the same local network.	Used in offices to connect many desktop computers.
Access Point (AP)	Extends or creates a wireless Wi-Fi network.	Installed in schools, hotels, malls, or large homes for better Wi-Fi coverage.
Network Cable (Ethernet)	Transfers data through a wired connection between devices.	Used to connect a PC, switch, router, or gaming console for a stable connection.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Cable Type	Common Use	Example Source
Ethernet (Cat5e/Cat6)	Used to connect computers, routers, switches, and gaming consoles in homes and offices.	https://en.wikipedia.org/wiki/Ethernet_over_twisted_pair
Coaxial Cable	Used for cable TV, cable internet, and connecting cable modems.	https://en.wikipedia.org/wiki/Coaxial_cable
Fiber Optic Cable	Used for high-speed internet and long-distance communication networks.	https://en.wikipedia.org/wiki/Optical_fiber

4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	Router	Switch
Main Use	Connects different networks (LAN to Internet).	Connects multiple devices within the same LAN.
Typical Location	Homes, offices, schools (connected to ISP).	Offices, computer labs, server rooms.
Data Handling	Uses IP addresses to forward packets between networks.	Uses MAC addresses to forward data within the local network.

Feature	Router	Switch
Internet Sharing	Yes, shares internet among devices.	No, only connects local devices unless connected to a router.
Wireless Support	Usually provides Wi-Fi.	Normally wired only (unless it's a special managed switch with wireless integration).
Security Features	Supports firewall, NAT, DHCP, VPN.	Basic switching; advanced managed switches offer VLANs and monitoring.